

Data Flow Diagram

Date	02-July-2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Description

The Smart Lender system follows a highly optimized, linear data flow to automate the credit underwriting process. The system begins when the external entity—the loan applicant—submits 11 independent financial and demographic indicators through our interactive layout interface (input.html).

This raw payload passes immediately into the central Flask Underwriting Routing Handler (app.py), where input data streams are validated and converted into floating-point vectors (float()) to filter out structural processing faults at the gate. Next, the array flows into the Dynamic Feature Standardization Module, which references the workspace file store to read scale1.pkl (the pre-trained Scikit-Learn StandardScaler binary). This scales values into uniform vector distributions, adjusting magnitudes before prediction execution.

The normalized vector array is then processed directly by the Automated Underwriting Classifier Module. This engine interacts with the serialized rdf.pkl binary to run multi-tree Random Forest calculations and output an instant categorical prediction score label (0 or 1).

Finally, the output vector triggers the Contextual Result Generator. The data-driven determination context is compiled via native Jinja2 conditional code loops inside output.html, dynamically loading the green credit approval console or red risk alert board to serve the final result back to the end user with maximum clarity.

